

SAQIB ALI

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PROFESSIONAL SUMMARY

A driven engineering student with a solid grounding in Real-time Control Systems and Signal Processing. Experienced in creating adaptive control solutions for high-inertia systems. Proficient in Fusion 360 for 3D modeling and design, and skilled in conducting Finite Element Analysis (FEA) to evaluate product performance and durability. Enthusiastic about emerging technologies, such as Generative AI and aerospace applications, and eager to apply technical skills in a challenging engineering position.

EDUCATION

Zakir Husain College of Engineering & Technology (ZHCET), Aligarh
B.Tech in Electrical Engineering | Sep 2023 - Jun 2027

Syedna Tahir Saifuddin (STS) School, AMU
Senior Secondary School (PCM)

PROFESSIONAL EXPERIENCE

Orbitalink Private Limited (in collaboration with AMU Innovation Foundation) |
Aligarh, UP
Electrical Design Intern

- Developed and implemented an Adaptive PID Controller designed to achieve stabilization and precise control of a high-inertia antenna tracking system.
- Applied advanced control theory principles, system modeling, and performance optimization for critical satellite communication infrastructure.

SS AMU SAT | Aligarh, UP

Ground station Team Member | Sep 2024 - Present

- Played a key role in developing and building the 3D model chassis for the antenna tracker.
- Conducted tests on antenna tracking mechanisms.
- Assisted with ground station operations and satellite tracking protocols.
- Helped maintain and calibrate communication equipment to guarantee reliable data transmission.
- Acquired hands-on experience in signal reception, hardware troubleshooting, and system diagnostics.

SKILLS

- **Hard Skills:** Real-time Control Systems, Signal Processing, Adaptive PID Control, System Modeling & Optimization.
- **Development:** Python, Full Stack Web Development, Data Science (NumPy, Pandas, Matplotlib, Seaborn), sensor interfacing, I2C protocols, C++ for hardware control.
- **Engineering Tools:** Aircraft Design Principles, Mechanical Product Design, FUSION 360, Control Block Design, Simulink.
- **Soft Skills:** Communication, Problem Solving, Adaptability, Team Collaboration, Detail-Oriented.

PROJECTS

Weather Dashboard Application

- Deployed on Render using OpenWeatherMap API, Node.js, and Axios.
- Developed a responsive weather forecasting app with Modern JavaScript and CSS3.
- Integrated OpenWeatherMap API for real-time data on temperature, humidity, and wind speed globally.
- Created a robust search function for querying weather conditions worldwide.
- Established error handling protocols to manage API failures and invalid inputs.

Simulink PID Control System Modeling

- Designed and simulated a dynamic control loop in Simulink to model the behavior of a physical plant.
- Verified controller performance against various setpoint changes and noise inputs.

Simulink Modeling and Analysis of a DC-DC Boost Converter

- Designed and simulated a DC-DC Boost Converter circuit in Simulink as part of my college laboratory work, showcasing the ability to step up a DC input voltage.
- Analyzed the converter's steady-state and transient responses under varying duty cycles and load conditions.
- This project demonstrates proficiency in Power Electronics, Circuit Simulation, and Switch-Mode Power Supply (SMPS) fundamentals.

3D Chassis Design for Motorized Tracking Unit

- Crafted the 3D model of the chassis for the motorized tracking unit using Fusion 360, optimizing the design for rigidity and minimal weight while ensuring proper placement for the motor, gearbox, and sensors.
- Implemented the core PID control logic for the two-axis antenna positioning system, which is crucial for maintaining a lock on a satellite signal.

CERTIFICATIONS & ACTIVITIES

- **Google AI Training:** Certifications in Develop GenAI Apps with Gemini and Streamlit, and Prompt Design in Vertex AI (May 2024).
- **IEEE Student Branch, AMU:** Certificate of Participation in MATLAB Simulation and Control (Oct 2025); Completed Introduction to Cybersecurity.
- **Advanced Aircraft Design Workshop:** Workshop on aircraft design (Bharat Space Education Research Centre, 2024).
- **SS AMU SAT:** Certificate of Participation (2024).
- **Indian Mobile Congress:** Represented Orbitalink and showcased the Field Unit (Oct 31, 2024).